

Project 1: End-to-End CI/CD Pipeline

Objective

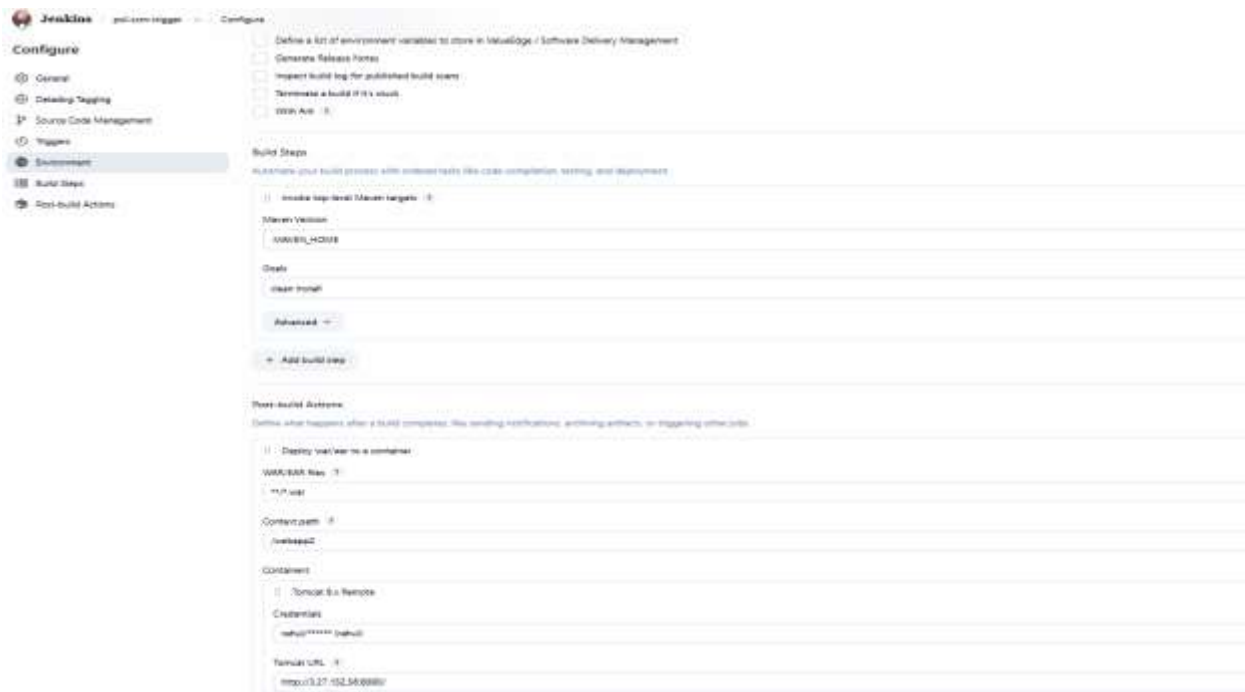
To implement a complete CI/CD pipeline where a Java web application is pushed to GitHub, built using Maven, automatically triggered by Jenkins, deployed to Tomcat, and verified via browser.

Process Description

1. Created GitHub repository and pushed Java application source code.
2. Configured Jenkins Freestyle Job with SCM Polling trigger.
3. Built the project using Maven (mvn clean package).
4. Generated WAR file inside target directory.
5. Deployed WAR file to Tomcat webapps directory.
6. Verified application in browser.

Result

Application deployed successfully and changes were reflected automatically after modifying pom.xml file.



Success build and deploy to tomcat

[illegible]

Git commands to change something and add to central repository

```
aws [Search] [Alt+S] Ask Amazon Q

Please make sure you have the correct access rights
and the repository exists.
[root@ip-172-31-18-182 webapp]# git push origin master
Username for 'https://github.com': ragulhb24@gmail.com
Password for 'https://ragulhb24@gmail.com@github.com':
To https://github.com/Rahul-max-dot/cab-booking.git
 ! (rejected)      master -> master (fetch first)
error: failed to push some refs to 'https://github.com/Rahul-max-dot/cab-booking.git'
hint: Updates were rejected because the remote contains work that you do not
hint: have locally. This is usually caused by another repository pushing to
hint: the same ref. If you want to integrate the remote changes, use
hint: 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
[root@ip-172-31-18-182 webapp]# git push origin master --force
Username for 'https://github.com': rahulhb24@gmail.com
Password for 'https://rahulhb24@gmail.com@github.com':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 3.95 KiB | 3.96 MiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Rahul-max-dot/cab-booking.git
 + 59afcff...90f2f3b master -> master (forced update)
[root@ip-172-31-18-182 webapp]# pwd
/var/www/html/cab-booking/taxi-booking/src/main/webapp
[root@ip-172-31-18-182 webapp]#
```

Change in pom.xml file

```
<ul>
    <li class="active"><a href="index.html">Home</a></li>
    <li><a href="services.html">Services at home</a></li>
    <li><a href="drivers.html">Drivers</a></li>
    <li><a href="cars.html">Cars</a></li>
    <li><a href="garage.html">Garage</a></li>
    <li><a href="locations.html">Locations</a></li>
    <li><a href="#">Pages</a>
        <ul>
            <li><a href="about.html">About</a></li>
            <li><a href="testimonials.html">Testimonials</a></li>
            <li><a href="jobs.html">Jobs</a></li>
            <li><a href="blog.html">Blog</a></li>
            <li><a href="contact.html">Contact</a></li>
            <li><a href="404.html">Error 404</a></li>
            <li><a href="#">Third Level Nav</a>
                <ul>
```

Result of after deploy to tomcat server

Applications					
Path	Version	Display Name	Warning	Access	Commands
/	None specified	Welcome to Tomcat	None	0	Start Stop Reload Undeploy Engine sessions with idle > 30 minutes
/docs	None specified	Tomcat Documentation	None	0	Start Stop Reload Undeploy Engine sessions with idle > 30 minutes
/examples	None specified	Servlet and JSP Examples	None	0	Start Stop Reload Undeploy Engine sessions with idle > 30 minutes
/html	None specified	Webapp01	None	0	Start Stop Reload Undeploy Engine sessions with idle > 30 minutes
/html2	None specified		None	0	Start Stop Reload Undeploy Engine sessions with idle > 30 minutes
/html3	None specified	Webapp01	None	0	Start Stop Reload Undeploy Engine sessions with idle > 30 minutes
/html4	None specified	Archetype Created Web Application	None	0	Start Stop Reload Undeploy Engine sessions with idle > 30 minutes
/html5	None specified		None	0	Start Stop Reload Undeploy Engine sessions with idle > 30 minutes
/html6	None specified	Tomcat Host Manager Application	None	0	Start Stop Reload Undeploy Engine sessions with idle > 30 minutes
/html7	None specified	Tomcat Manager Application	None	1	Start Stop Reload Undeploy Engine sessions with idle > 30 minutes
/test/html5.1.1	None specified	Webapp01	None	0	Start Stop Reload Undeploy Engine sessions with idle > 30 minutes

Final output of the project (changed from “sevices” to “services at home”)



Project 2: Automated Application Deployment using Ansible

Objective

To automate Java application deployment on Tomcat servers using Ansible playbooks.

Process Description

1. Created Ansible inventory for control and target nodes.
2. Installed Java, Maven, and Tomcat using Ansible playbook.
3. Cloned GitHub repository automatically.
4. Built WAR file using Maven.
5. Copied WAR file to Tomcat webapps directory.
6. Restarted Tomcat service using Ansible.
7. Validated deployment via browser.

Result

Application deployment completed successfully without manual intervention.

```
---
- name: Install Maven
  hosts: all
  become: yes

  tasks:
    - name: Download Maven
      get_url:
        url: https://d1cdn.apache.org/maven/maven-3/3.9.12/binaries/apache-maven-3.9.12-bin.tar.gz
        dest: /opt/

    - name: Extract Maven
      unarchive:
        src: /opt/apache-maven-3.9.12-bin.tar.gz
        dest: /opt/
        remote_src: yes

    - name: removing old file
      file:
        path: /opt/apache-maven-3.9.12-bin.tar.gz
        state: absent

    - name: install java
      yum:
        name: java
        state: present
    - name: Install Java 17
      yum:
        name: java-17-amazon-corretto
        state: present
```

Tomcat setup

- name: Configure Tomcat Manager

hosts: tomcat

become: yes

vars:

tomcat_dir: /opt/apache-tomcat-10.1.52

tasks:

- name: Configure tomcat-users.xml

copy:

dest: /opt/apache-tomcat-10.1.52/conf/tomcat-users.xml"

content: |

<?xml version="1.0" encoding="UTF-8"?>

<tomcat-users>

<role rolename="manager-gui"/>

<role rolename="manager-script"/>

<user username="tomcat" password="tomcat" roles="manager-gui"/>

<user username="rahul" password="rahul" roles="manager-script"/>

</tomcat-users>

- name: Remove Manager IP restriction

copy:

dest: /opt/apache-tomcat-10.1.52/webapps/manager/META-INF/context.xml

content: |

<Context privileged="true">

</Context>

- name: Restart Tomcat

shell: |

/opt/apache-tomcat-10.1.52/bin/shutdown.sh || true

sleep 5

/opt/apache-tomcat-10.1.52/bin/startup.sh

Tomcat direct deployment through ansible-playbook

- name: Clone and Deploy to Tomcat

hosts: jenkins

become: yes

vars:

repo_url: "https://github.com/Rahul-max-dot/Java-Web-CongratsApp.git"

app_name: "myapp"

deploy_path: /opt

tomcat_dir: /opt/apache-tomcat-10.1.52

tasks:

- name: Install git and maven (RHEL/CentOS)

yum:

name:

- git

- maven

state: present

- name: Clone repository

git:

repo: https://github.com/Rahul-max-dot/Java-Web-CongratsApp.git

dest: /opt/congrates.app

version: main

force: yes

- name: Build WAR file

shell: mvn clean package

args:

chdir: /opt/congrates.app

- name: Fetch WAR to control node

fetch:

src: /opt/congrates.app/target/webhook-trigger.war

dest: /tmp/webhook-trigger.war

```
/opt/apache-tomcat-10.1.52/bin/startup.sh
```

```

"msg": "Failed to connect to the host via ssh: Missing: Permanently added '172.31.14.28' (SSH25519) to the list of known hosts./bin/ssh172.31.14.28: permission denied (publickey,gssapi-keyex,gssapi-with-mic,password).",
  "unreachable": true
}

C:\Users\user>ssh -o StrictHostKeyChecking=no 172.31.14.28
root@172-31-13-63:~# su anadsta
anadsta@172-31-13-63:~$ cd
anadsta@172-31-13-63:~$ ansible all -m ping
WARNING: Platform linux on host 172.31.14.28 is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of another Python interpreter could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
172.31.14.28 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": false,
  "ping": "pong"
}

WARNING: Platform linux on host 172.31.8.214 is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of another Python interpreter could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
172.31.8.214 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": false,
  "ping": "pong"
}

anadsta@172-31-13-63:~$

```



```

TASK [Build WAR file] *****
changed: [172.31.14.28]
changed: [172.31.8.214]

TASK [Fetch WAR to control node] *****
changed: [172.31.14.28]
ok: [172.31.8.214]

PLAY [Deploy to Tomcat Servers] *****

TASK [Gathering Facts] *****
ok: [172.31.14.28]
ok: [172.31.8.214]

TASK [Copy WAR to Tomcat] *****
changed: [172.31.14.28]
changed: [172.31.8.214]

TASK [Restart Tomcat] *****
changed: [172.31.14.28]
changed: [172.31.8.214]

PLAY RECAP *****
172.31.14.28      : ok=4    changed=6    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
172.31.8.214     : ok=4    changed=5    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

[asadmin@ip-172-31-13-63 ansible]$

```

Tomcat Web Application Manager

Message: OK

Manager			
List Applications	HTML Manager Help	Manager Help	Server Status

Applications					
Path	Version	Display Name	Running	Sessions	Commands
/	None specified	Welcome to Tomcat	true	0	Start Stop Reload Undeploy Expire sessions with idle < 30 minutes
/docs	None specified	Tomcat Documentation	true	0	Start Stop Reload Undeploy Expire sessions with idle < 30 minutes
/examples	None specified	Servlet and JSP Examples	true	0	Start Stop Reload Undeploy Expire sessions with idle < 30 minutes
/host-manager	None specified	Tomcat Host Manager Application	true	0	Start Stop Reload Undeploy Expire sessions with idle < 30 minutes
/manager	None specified	Tomcat Manager Application	true	1	Start Stop Reload Undeploy Expire sessions with idle > 30 minutes
/archbook/00001	None specified	Archetype Created Web Application	true	0	Start Stop Reload Undeploy Expire sessions with idle < 30 minutes

Deploy
Deploy directory or WAR file located on server



Project 3: CI + CD using Jenkins and Ansible

Objective

To integrate Jenkins and Ansible for fully automated Continuous Integration and Continuous Deployment.

Process Description

1. Jenkins built the application using Maven.
2. Jenkins triggered Ansible playbook after successful build.
3. Ansible deployed WAR file to Tomcat server.
4. Tomcat restarted automatically.
5. Application verified successfully in browser.

Configure

- General
- Datadog Tagging
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-build Actions

General

Enabled 

Description

Plain text [Preview](#)

Site

Datadog Tagging

- ☐ Add tags from file in workspace
- ☐ Add tags from list of properties

Advanced 

- ☐ Discard old builds 
- ☐ Enable Datadog Test Optimization 
- ☐ GitHub project
- ☐ This project is parameterised 
- ☐ Throttle build 
- ☐ Execute concurrent build if necessary 

Advanced 

Source Code Management

Connect and manage your code repository to automatically pull the latest code for your builds.

- ☐ None
- ☐ Bitbucket Server
- ☒ Git 

Repositories 

Repository URL 

<https://github.com/Rahul-nai-dot-lava-Web-CongratApp.git>

Credentials ?

- none -

+ Add

Advanced ▾

+ Add Repository

Branches to build ?

Branch Specifier (blank for 'any') ?

* / main

+ Add Branch

Repository browser ?

(Auto)

Additional Behaviours

+ Add

Triggers

Set up automated actions that start your build based on specific events, like code changes or scheduled times.

- ☐ Trigger builds remotely (e.g. from scripts) ?
- ☐ Bitbucket webhook trigger ?
- ☐ Build after other projects are built ?
- ☐ Build periodically ?
- ☒ GitHub hook trigger for GITScm polling ?
- ☐ Poll SCM ?

Environment

Configure settings and variables that define the context in which your build runs, like credentials, paths, and global parameters.

- ☐ Delete workspace before build starts
- ☐ Use secret text(s) or file(s) ?
- ☐ Provide Configuration files ?
- ☐ Add timestamps to the Console Output

Invoke top-level Maven targets ?

Maven Version

MAVEN_HOME

Goals

clean install

Advanced

+ Add build step

Post-build Actions

Define what happens after a build completes, like sending notifications, archiving artifacts, or triggering other jobs.

Deploy war/ear to a container

WAR/EAR files ?

**/*.war

Context path ?

/home21

Containers

Tomcat 5.x Remote

Credentials

rahu/***** (rahu)

Tomcat URL ?

http://3.107.239.52:8080/

Advanced

+ Add Container

☐ Deploy on failure

+ Add post-build action

Deployment to tomcat

```
Started by GitHub push by Rahul-mas-dot
Running as SYSTEM
[ATLASIAN CLOUD PLUGIN] [WARN] Not a WorkflowRun, automatic builds and deployments won't work.
Building in workspace /var/lib/jenkins/workspace/webhook-trigger
The recommended git tool is: NONE
No credentials specified
> git rev-parse --resolve-git-dir /var/lib/jenkins/workspace/webhook-trigger/.git # timeout=10
Fetching changes from the remote Git repository
> git config remote.origin.url https://github.com/Rahul-mas-dot/Java-Web-CongratsApp.git # timeout=10
Fetching upstream changes from https://github.com/Rahul-mas-dot/Java-Web-CongratsApp.git
> git --version # timeout=10
> git --version # 'git version 2.50.1'
> git fetch --tags --force --progress -- https://github.com/Rahul-mas-dot/Java-Web-CongratsApp.git +refs/heads/*:refs/remotes/origin/* # timeout=30
> git rev-parse refs/remotes/origin/main^{commit} # timeout=10
Checking out Revision 0002826e40702ab3e9f7ac7f8fab43275479506 (refs/remotes/origin/main)
> git config core.sparsecheckout # timeout=10
> git checkout -f 0002826e40702ab3e9f7ac7f8fab43275479506 # timeout=10
Commit message: "Update index.jsp"
> git rev-list --no-walk bc73ccae4474bb38b20d8ac2e1a520924b6cbb # timeout=10
[webhook-trigger] $ /usr/share/maven/bin/mvn clean install
[INFO] Scanning for projects...
[INFO]
[INFO] ----- com.webhook-trigger > -----
[INFO] Building webhook-trigger 1.0.0
[INFO] -----[ war]-----
[INFO]
[INFO] --- maven-clean-plugin:2.5:clean (default-clean) @ webhook-trigger ---
[INFO] Deleting /var/lib/jenkins/workspace/webhook-trigger/target
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ webhook-trigger ---
[INFO] Using 'UTF-8' encoding to copy filtered resources.
[INFO] skip non existing resourceDirectory /var/lib/jenkins/workspace/webhook-trigger/src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.11.0:compile (default-compile) @ webhook-trigger ---
[INFO] No sources to compile
[INFO]
[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ webhook-trigger ---
[INFO] Using 'UTF-8' encoding to copy filtered resources.
[INFO] skip non existing resourceDirectory /var/lib/jenkins/workspace/webhook-trigger/src/test/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.11.0:testCompile (default-testCompile) @ webhook-trigger ---
[INFO] No sources to compile
[INFO]
[INFO] --- maven-surefire-plugin:2.12.4:test (default-test) @ webhook-trigger ---
[INFO] No tests to run.
[INFO]
[INFO] --- maven-war-plugin:3.3.2:war (default-war) @ webhook-trigger ---
[INFO] Packaging webapp [webhook-trigger] in [/var/lib/jenkins/workspace/webhook-trigger/target/webhook-trigger]
[INFO] Processing war project
[INFO] Copying webapp resources [/var/lib/jenkins/workspace/webhook-trigger/src/main/webapp]
[INFO] Building war: /var/lib/jenkins/workspace/webhook-trigger/target/webhook-trigger.war
[INFO]
[INFO] --- maven-install-plugin:2.4:install (default-install) @ webhook-trigger ---
[INFO] Installing /var/lib/jenkins/workspace/webhook-trigger/target/webhook-trigger.war to /var/lib/jenkins/.m2/repository/com/webhook/webhook-trigger/1.0.0/webhook-trigger-1.0.0.war
[INFO] Installing /var/lib/jenkins/workspace/webhook-trigger/pom.xml to /var/lib/jenkins/.m2/repository/com/webhook/webhook-trigger/1.0.0/webhook-trigger-1.0.0.pom
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 0.802 s
[INFO] Finished at: 2025-02-17T13:45:00Z
[INFO] -----
[DeployPublisher][INFO] Attempting to deploy 1 war file(s)
[DeployPublisher][INFO] Deploying /var/lib/jenkins/workspace/webhook-trigger/target/webhook-trigger.war to container Tomcat 9.x Remote with context /home21
  Redeploying [/var/lib/jenkins/workspace/webhook-trigger/target/webhook-trigger.war]
  Undeploying [/var/lib/jenkins/workspace/webhook-trigger/target/webhook-trigger.war]
  Deploying [/var/lib/jenkins/workspace/webhook-trigger/target/webhook-trigger.war]
[ATLASIAN CLOUD PLUGIN] [WARN] Not a WorkflowRun, onCompleted() won't be propagated to listeners
Finished: SUCCESS
```

```
awsadmin ec2-user
[root@ip-172-31-14-28 home]# cd
[root@ip-172-31-14-28 ~]# ls /home/
awsadmin ec2-user
[root@ip-172-31-14-28 ~]# cd /opt/
[root@ip-172-31-14-28 opt]# ls
apache-maven-3.9.12 apache-tomcat-10.1.52 aws congrates.app
[root@ip-172-31-14-28 opt]# cd congrates.app/
[root@ip-172-31-14-28 congrates.app]# ls
README.md pom.xml src target
[root@ip-172-31-14-28 congrates.app]# cd target/
[root@ip-172-31-14-28 target]# ls
maven-archiver webbook-trigger webbook-trigger.war
[root@ip-172-31-14-28 target]# cd
[root@ip-172-31-14-28 ~]# ls
[root@ip-172-31-14-28 ~]# cd /opt/
[root@ip-172-31-14-28 opt]# ls
apache-maven-3.9.12 apache-tomcat-10.1.52 aws congrates.app
[root@ip-172-31-14-28 opt]# cd apache-tomcat-10.1.52/
[root@ip-172-31-14-28 apache-tomcat-10.1.52]# ls
NOTICE.txt CONTRIBUTING.md LICENSE NOTICE README.md RELEASE-NOTES RUNNING.txt bin conf lib logs temp webapps work
[root@ip-172-31-14-28 apache-tomcat-10.1.52]# cd webapps/
[root@ip-172-31-14-28 webapps]# ls
ash: ls: command not found
[root@ip-172-31-14-28 webapps]# ls
COPY docs examples host-manager manager webbook-trigger webbook-trigger.war
[root@ip-172-31-14-28 webapps]#
```

i-0c85e061b4b71cb1e (slave-server)
PublicIP: 3.27.244.123 PrivateIP: 172.31.14.28

Tomcat Web Application Manager

Message: OK

Manager

List Applications

HTML Manager Help

Manager Help

Server Status

Path	Version	Display Name	Running	Sessions	Commands
/	None specified	Welcome to Tomcat	true	0	Start Stop Reload Undeploy Expire sessions: with idle > 30 minutes
/docs	None specified	Tomcat Documentation	true	0	Start Stop Reload Undeploy Expire sessions: with idle > 30 minutes
/examples	None specified	Servlet and JSP Examples	true	0	Start Stop Reload Undeploy Expire sessions: with idle > 30 minutes
/host-manager	None specified	Tomcat Host Manager Application	true	0	Start Stop Reload Undeploy Expire sessions: with idle > 30 minutes
/manager	None specified	Tomcat Manager Application	true	1	Start Stop Reload Undeploy Expire sessions: with idle > 30 minutes
/webbook-trigger	None specified	Archetype Created Web Application	true	0	Start Stop Reload Undeploy Expire sessions: with idle > 30 minutes

Deploy

Deploy directory or WAR file located on server

Context Path: /

Output of webook-trigger

Congratulations You Have Successfully Deployed Java Applications with version - 30



Congratulations You Have Successfully Deployed Java Applications with version - 50



Project 4: Git Branching and Release Management

- **Organize development work:** By separating *main*, *develop*, and *feature* branches, you avoid chaos and keep production-ready code safe.

- **Enable collaboration:** Pull Requests allow peer review, discussion, and controlled integration of new features.
- **Handle conflicts gracefully:** Learning to resolve merge conflicts ensures smooth teamwork when multiple people edit the same files.
- **Mark stable releases:** Tagging versions (like `v1.0`) creates clear checkpoints in history, making it easy to roll back or reference specific releases.

```

(root@ip-172-31-13-63 release-manager) # git checkout -b main
fatal: a branch named 'main' already exists
(root@ip-172-31-13-63 release-manager) # git checkout -b develop
switched to a new branch 'develop'
(root@ip-172-31-13-63 release-manager) # git checkout -b feature/login
switched to a new branch 'feature/login'
(root@ip-172-31-13-63 release-manager) # git init
Initialized empty Git repository in /root/.release-manager/.git/
(root@ip-172-31-13-63 release-manager) # git add index.html
(root@ip-172-31-13-63 release-manager) # git commit -m "add login features"
[feature/login a09e574] add login features
Committer: root <root@ip-172-31-13-63.ap.southwest-2.compute.internal>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

    git config --global --edit

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

1 file changed, 1 insertion(+)
create mode 100644 index.html
(root@ip-172-31-13-63 release-manager) #

```

i-0c745180c548ba5a1 (ansible-node)

PublicPc: 16.176.206.24 PrivatePc: 172.31.13.63

```

switched to branch 'main'
Your branch is up to date with 'origin/main'.
(root@ip-172-31-13-63 release-manager) # git merge develop
Updating files64b..256a794
Fast-forward
 index.html | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 index.html
(root@ip-172-31-13-63 release-manager) # git tag v1.0
(root@ip-172-31-13-63 release-manager) # git push origin v1.0
Username for 'https://github.com': rahulhb24@gmail.com
Password for 'https://rahulhb24@gmail.com:github.com':
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 322 bytes | 312.00 KiB/s, done.
Total 3 deltas 0, reused 0 deltas 0, pack-reused 0 (from 0)
to https://github.com/rahul-max-dot-release-manager.git
 * [new tag]         v1.0 -> v1.0
(root@ip-172-31-13-63 release-manager) # git log --online
fatal: unrecognized argument: --online
(root@ip-172-31-13-63 release-manager) # git log --oneline
256a794 (HEAD -> main, tag: v1.0, develop) resolved conflict
a09e574 (feature/login, feature/login) add login features
12ac749 (origin/main, origin/HEAD) initial commit
(root@ip-172-31-13-63 release-manager) #

```

i-0c745180c548ba5a1 (ansible-node)

PublicPc: 16.176.206.24 PrivatePc: 172.31.13.63

